NESS-Net: An NAMLab Edge-Guided and Scribble-Supervised Swin-Transformer Net for RGB-D Salient Object Detection

Yunping Zheng, Hao Li, Shiqiang Shu, Ke Xu, and Mudar Sarem

***Abstract*—The advent of scribble-supervised learning has opened new frontiers in weakly supervised RGB-D salient object detection (SOD), representing a novel paradigm that significantly reduces annotation costs while maintaining detection efficacy. The SOD methods predominantly based on convolutional neural networks (CNNs) exhibit notable limitations in capturing global contextual dependencies and effectively utilizing hierarchical multiscale representations. Furthermore, conventional edge extraction approaches based on Canny edge detection demonstrate inherent limitations in preserving structural coherence across diverse object boundaries. To address these critical challenges, we propose an NESS-Net model, an innovative weakly supervised architecture that synergistically integrates our previously developed NAMLab framework with Swin-Transformer-based feature learning. The core contributions of our work are threefold: Firstly, building upon our prior work in perceptual-aware image segmentation, we employ the NAMLab algorithm—a hierarchical segmentation framework inspired by Nonsymmetry and Anti-packing pattern representation Model in the Lab color space (NAMLab)—to generate edge maps that better align with human visual perception. Secondly, we devise a U-shaped Edge Construction Module (ECM) that systematically refines edge features through progressive refinement of the NAMLab-generated edge priors. Thirdly, leveraging Swin-Transformer's hierarchical attention mechanism, we put forward a lightweight Cross-Attention Fusion Module (CAFM) that establishes long-range dependencies across modalities while maintaining computational efficiency. This architectural innovation naturally extends to our Cross-Modal multi-scale Attention-weighting Module (CMAM), which explicitly models inter-modal relationships through transformer-based attention weighting. The extensive experimental results on nine benchmark RGB-D SOD datasets demonstrate that our proposed NESS-Net model outperforms all the current state-of-the-art scribble-supervised models and even achieves performance competitive to leading fully supervised models. The source code and pre-trained models will be made publicly available at https://github.com/EricLie-c/NESS-Net**

***Index Terms*—Weakly-supervised, scribble-supervised, RGB-D salient object detection, cross-modality feature fusion, NAMLab, Swin-Transformer, cross-attention fusion.**

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I. Introduction

THE human visual system demonstrates remarkable efficiency in attentional prioritization within complex visual environments, a capability attributed to its specialized neural architecture and perceptual selection mechanisms. This biological capability has inspired significant advancements in computational modeling through RGB-based salient object detection (SOD). However, the inherent limitations of unimodal spectral information in capturing comprehensive scene semantics have prompted the computer vision community to explore multimodal integration strategies. The emergence of RGB-D SOD represents a paradigm shift in visual processing, where depth information provides critical geometric cues to enhance foreground-background differentiation, particularly in challenging scenarios characterized by cluttered environments or low-contrast conditions.

Despite the impressive performance of fully supervised RGB-D SOD methods, their heavy reliance on large-scale, densely annotated datasets [1], [2], [3] incurs significant labor costs. To mitigate this, many previous studies have explored weakly supervised training strategies using sparse annotations, such as class labels [4], [5], bounding boxes [6], or scribbles [7], [8]. While scribbles are cost-effective, they inherently lack detailed information regarding global internal structures and precise object boundaries. One approach to compensate for this limitation is to incorporate prior cues from traditional techniques, such as extracting edge or structural information from input images [9], [10]. For instance, Xu et al. [9] employed the Canny algorithm for edge extraction and adopted self-training methods to refine manually enhanced scribbles. However, the edge features derived from these strategies are often inadequate. Many existing works, such as [10], rely on traditional CNN models (e.g., ResNet [11] and VGG [12]), which lead to insufficient modeling of global long-range dependencies. Moreover, these studies frequently overlook the importance of multi-level semantic extraction and fusion. For example, the authors in [9] utilized outputs from only certain stages of the backbone network to extract edge features, inevitably resulting in the loss of critical semantic information and lower-quality predictions in difficult scenarios.

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To overcome the limitations of traditional CNNs in capturing global context, the Swin-Transformer [13] has emerged as a groundbreaking architecture, achieving state-of-the-art performance by effectively modeling long-range dependencies. However, how to synergize such powerful semantic learners with structural priors remains an open question. Our previous work in 2021 [14] introduced a hierarchical image segmentation framework based on the Nonsymmetry and Anti-packing pattern representation Model in the Lab color space (NAMLab). While the NAMLab excels at preserving structural coherence and generating "meaningful" regions consistent with human perception, it remains a traditional segmentation tool that lacks high-level semantic guidance. There is a clear research gap in effectively integrating the structural precision of the NAMLab with the semantic representation of Swin-Transformer to address the boundary-missing issue in scribble-supervised tasks. The NAMLab relies on low-level structural cues and does not explicitly model semantic saliency, which may lead to incomplete object boundaries or confusion under complex backgrounds. When depth maps are noisy or inconsistent, purely structural priors may become unreliable and can introduce errors if fused without adaptive weighting. In a word, an NAMLab alone cannot bridge the gap between sparse scribbles and dense supervision required for accurate saliency prediction.

Therefore, we propose an NAMLab Edge-Guided and Scribble-Supervised Swin-Transformer Net (NESS-Net). We transition from the traditional framework of [14] to a deep learning paradigm, utilizing the Swin-Transformer as a backbone. To bridge the gap between sparse scribbles and precise boundaries, we design the Edge Construction Module (ECM), which integrates the NAMLab-derived edges as an offline edge prior. Unlike [9], our ECM constructs multi-level edge representations via a U-shaped cascaded mechanism, enabling semantically consistent boundary learning and alleviating boundary-missing issues under scribble supervision. Furthermore, we introduce the Cross-Attention Fusion Module (CAFM) and the Cross-Modal multi-scale Attention-weighting Module (CMAM) to achieve efficient and interpretable cross-modal integration. The CAFM performs efficient cross-modal interaction using local window proposals, which reduces the computational overhead of global cross-attention while maintaining effective fusion of RGB and depth features. The CMAM adaptively re-calibrates the contributions of RGB and depth streams at multiple scales, suppressing unreliable depth cues under noisy depth or complex backgrounds and improving robustness. The main contributions of this paper are summarized as follows:

1) We put forward an effective NAMLab-based Edge Construction Module (ECM), which utilizes the NAMLab boundary maps and a U-Net [15] style cascaded sampling mechanism to guide the encoder in extracting and constructing the boundary features.
2) Inspired by the cross-attention mechanism, we propose a Cross-Attention Fusion Module (CAFM) based on the local window proposals, which can effectively learn richer contextual information and grasp the common factors between modalities.
3) We put forward a Cross-Modal multi-scale Attention-weighting Module (CMAM) that integrates the CAFM, extracting features in a pyramid pooling-like manner, providing interpretability for the weight analysis between different modalities.
4) The experimental results on nine public benchmark datasets show that our proposed NESS-Net model outperforms the state-of-the-art open-source scribble-based RGB-D SOD models, and surpasses many previous SOTA fully supervised RGB-D SOD methods, even exceeding the current most advanced fully supervised methods on some datasets.

II. Related Works

A. RGB-D Salient Object Detection

Early researches primarily focused on locating salient objects through regional contrast or low-level visual cues, but these methods often neglected the refined modeling of target boundaries, resulting in edge blurring or false detection issues [16]. To address this, deep learning-based SOD methods have achieved remarkable performance by exploring boundary cues and feature embedding [17], [18], often employing edge detection as an auxiliary task to enhance sensitivity to target contours [7]. Building on this, most existing RGB-D SOD models emphasize effective multimodal feature fusion [19], [20], [21] by leveraging additional depth maps to supply scene-specific geometric cues. This fusion can be implemented via implicit multimodal feature aggregation or through explicit assessment of modal contributions [22]. Given that the depth information is often a subject to noise, a key research focus has been on investigating the depth contribution [23], [24], [25] to extract more robust and informative depth data for multimodal learning. For example, Wang and Gong [25] introduced a dual module designed to fuse the noise-corrected depth maps with the RGB features, whereas Fan et al. [26] developed a depth depurator unit (DDU) and a three-stream feature learning module (FLM), which performed low-quality depth map filtering and cross-modal feature learning. Similar objectives have also been pursued with the incorporation of auxiliary depth estimation modules [27] or refinement modules that enhanced the geometric information utilization through self-supervised learning [28].

B. Weakly Supervised Salient Object Detection

In the field of weakly supervised salient object detection, the main types of weak annotations include image or category-level labels [28], [29], salient object subitizing [30], hand-drawn scribbles [7], [8], [9], bounding boxes [6], and noisy labels derived from the traditional handcrafted feature methods [31], [32], [33]. By recognizing the intrinsic link between gaze prediction and SOD, Zhao et al. [34] employed gaze maps as weak supervisory signals for video SOD, whereas Gao et al. [24] explored point supervision for video saliency detection. WBNet [35] introduced a multi-source framework that combines pseudo-background priors with point- and box-prompted SAM to generate more reliable supervision from sparse scribble labels. Moreover, Zhuge et al.

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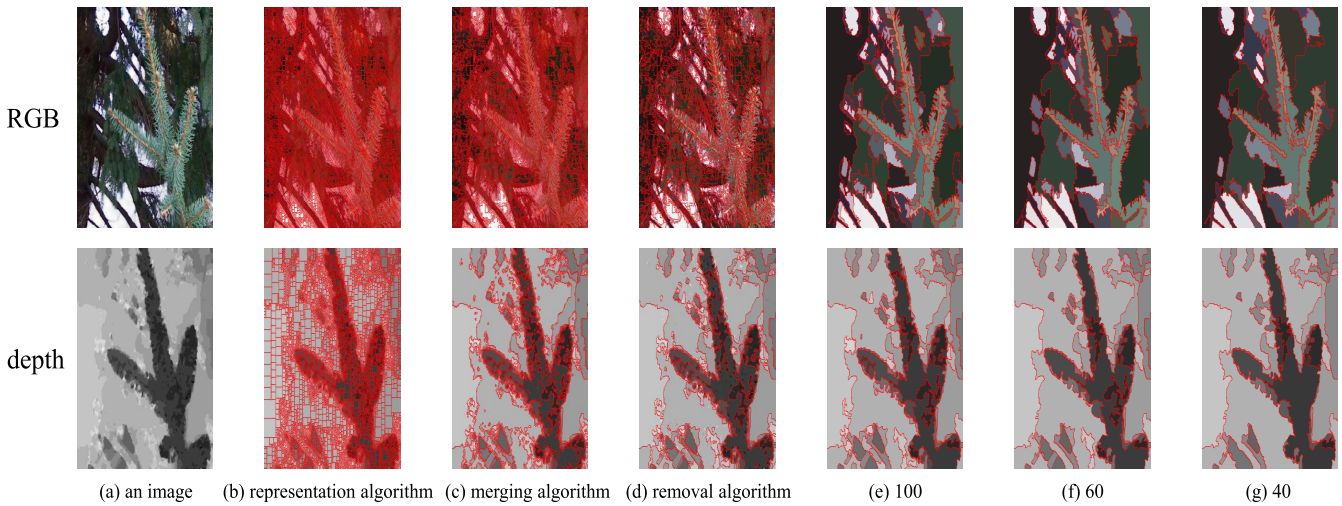


Fig. 1. (a) represents the original test image, (b) shows the results of the NAMLab-based color image representation algorithm, (c) shows the results of the NAMLab-based merging algorithm of homogeneous blocks, (d) presents the results of the NAMLab-based removal algorithm of residual regions, (e) to (g) respectively demonstrate the results with the NAMLab blocks counts of 100, 60, and 40.

[36] proposed to learn motion and temporal cues for unsupervised video object segmentation. By leveraging the class activation mapping techniques [37], the image or category-level supervision methods [23], [29] offered coarse foreground localization by highlighting the most discriminative regions of the objects. In contrast, the scribble supervision approaches [7], [8], [9], [10] provided precise yet sparse supervision. For instance, Xu et al. [9] pioneered the use of Dual Edge Detection modules (DEDs) to extract edge features for guiding salient object prediction. However, existing scribble-based methods still face limitations, such as the potential loss of multi-level semantics when using only specific backbone stages for edge detection, or the lack of feature richness when relying solely on prediction maps for edge supervision. Although the bounding box supervision [6] generally covers the entire salient foreground, it tends to include excessive background areas—especially for objects with non-convex shapes—leading to numerous false positives. Additionally, the techniques based on the noisy labels [33], [38] focused on extracting reliable supervisory information from the labels generated by the traditional methods.

C. NAMLab Edge Segmentation Algorithm

Inspired by the idea of the Nonsymmetry and Anti-packing pattern representation Model in the Lab color space (NAMLab) and the "global-first" invariant perceptual theory, in our previous research, we put forward a novel hierarchical image segmentation framework to effectively segment an image into regions that are "meaningful" to the human visual perception [14]. This framework redefined hierarchical image segmentation as a series of global optimization problems rather than sequential local decision tasks, thereby achieving precise localization of image boundaries.

For a more intuitive illustration of our algorithm, Fig. 1 presents the original images along with intermediate results produced by the NAMLab-based hierarchical image segmentation algorithm. In these figures, the edges of the segmented regions are delineated in red. The experimental results indicate that our proposed algorithm not only preserves more intricate details of object boundaries but also more effectively identifies foreground objects with similar color distributions compared to state-of-the-art methods. Moreover, our approach demonstrates faster execution speeds and reduced memory consumption.

III. Our Proposed NESS-Net for RGB-D SOD

A. The Overall Architecture of Our Proposed NESS-Net

In Fig. 2, our NAMLab edge-guided scribble-supervised network (NESS-Net) primarily comprises the following modules: Edge Construction Module (ECM), Cross-Attention Fusion Module (CAFM), and Cross-Modal multi-scale Attention-weighting Module (CMAM). The backbone network is based on Swin-Transformer [13], and to perceive more precisely salient objects in the space, channel-spatial attention modules containing Convolutional Block Attention Module (CBAM) [39] have been added to the backbone. To provide explicit geometric guidance, we leverage the NAMLab algorithm [14]—an existing hierarchical image segmentation tool—to generate initial edge maps from both RGB and depth images as structural priors. The backbone network extracts the RGB and depth images at different feature levels, denoted as R_i and D_i ($i \in [1, 5]$). These features, along with the NAMLab-derived edge cues, are subsequently incorporated into the corresponding cascaded sampling fusion modules and the Edge Construction Module (ECM), ultimately generating R'_2, R'_5 and D'_2, D'_5 and R_e, D_e, respectively. The R'_2, R'_5 and D'_2, D'_5 are finally fused with the refined edge features R_e and D_e, and then enter the CMAM with the corresponding RGB stream or depth stream features for modal weighting and concatenation, resulting in the preliminary feature S_1. Subsequently, S_1 continues to fuse with the edge features R_e

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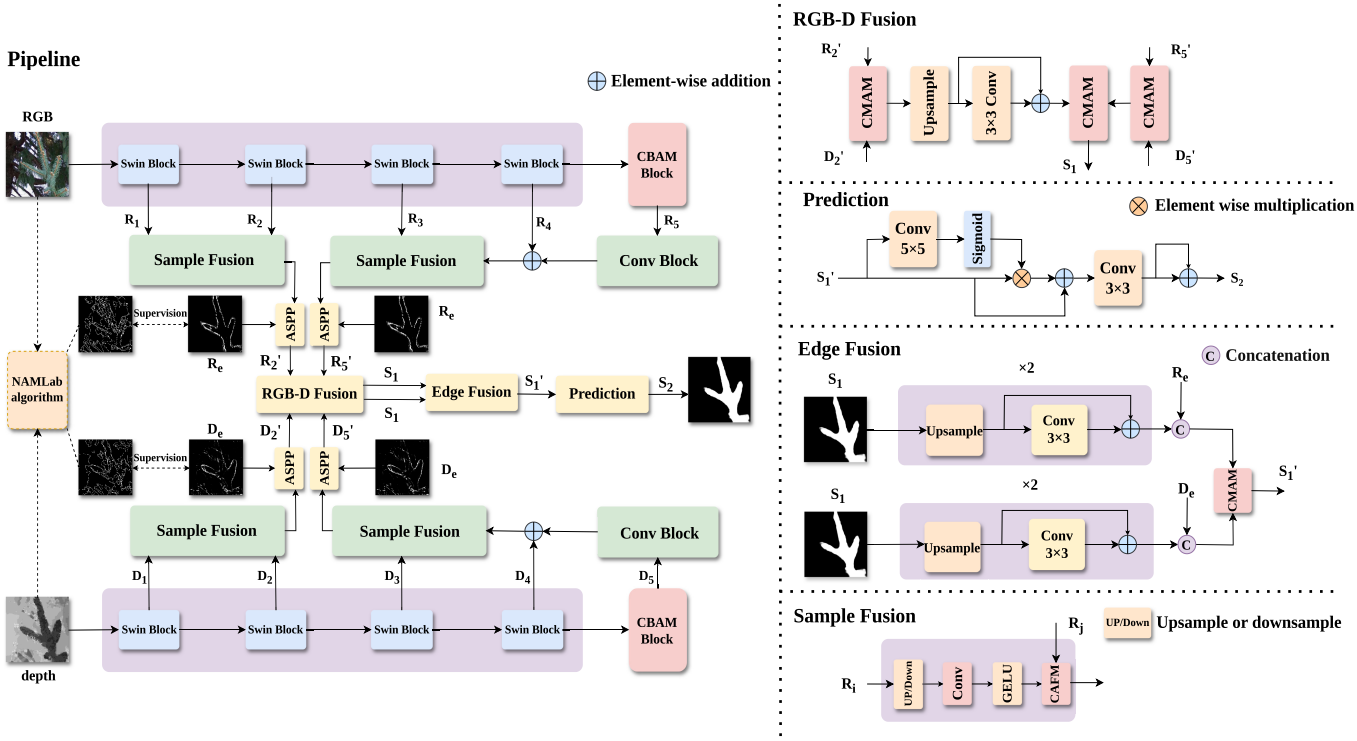


Fig. 2. The Overall Architecture of our proposed NESS-Net. Swin-Transformer is used as the backbone network for both the RGB stream and the depth stream. Meanwhile, R_e and D_e are generated from Edge Construct Module (ECM), which will be explained in Fig. 3. The internal structures of the Cross-Attention Fusion Module (CAFM) and the Cross-Modal multi-scale Attention-weighting Module (CMAM) will be explained in Fig. 4 and Fig. 6 respectively.

and D_e, and through a self-gated prediction module, ultimately produces the prediction map S_2.

B. Edge Construction Module (ECM)

Before delving into the implementation details, we clarify the motivation behind the Edge Construction Module (ECM). Under weak supervision settings like scribbles, networks often struggle to capture precise object boundaries due to label sparsity. The primary objective of the ECM is to bridge the gap between sparse annotations and complete boundary delineation. Inspired by the cascaded sampling structure of the U-Net [15], addressing the issue in the previous models in the field [9] that only select the outputs from specific stages for edge feature extraction, leading to incomplete edge features and loss of semantic information, we have proposed a U-Net style cascaded sampling module to construct more comprehensive edge semantic information.

The design of the Edge Construction Module (ECM) is intricately aligned with the hierarchical architecture of our backbone network. Specifically, our backbone consists of a standard Swin-Transformer followed by a Convolutional Block Attention Module (CBAM). The Swin-Transformer inherently generates multi-scale feature maps across four stages, capturing a spectrum of information ranging from low-level textures to high-level semantics. Supplementing this, the CBAM aggregates global context information, effectively establishing the fifth feature level.

We adopt this specific 5-level configuration based on two key rationales. Firstly, utilizing fewer levels (e.g., only the top 3) would sacrifice critical shallow spatial details required for crisp boundary delineation or lack sufficient global context to suppress non-salient edges. Secondly, expanding to more levels (e.g., 7) would necessitate artificial interpolation, introducing computational redundancy without genuine information gain given the fixed depth of the backbone.

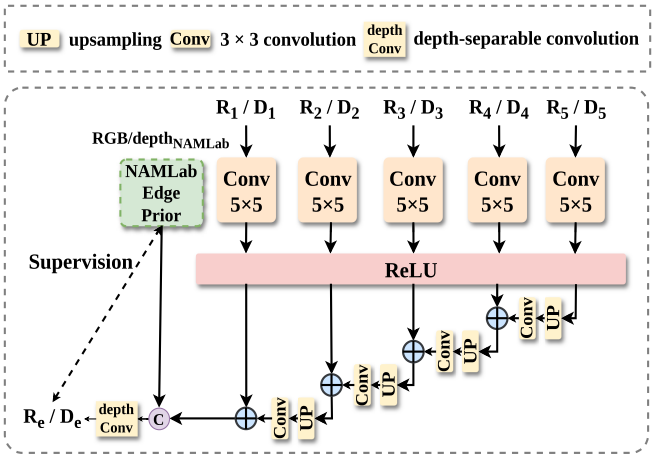


Fig. 3. Illustration of the internal structure of our Edge Construction Module (ECM).

Consequently, feature maps from these five levels serve as inputs to the ECM. To further enable the explicit structural cues from the NAMLab algorithm to guide the edge feature construction, the 1-channel NAMLab edge priors (RGB_{NAMLab} and $depth_{NAMLab}$) are concatenated with the decoded deep features at the end of the module. As illustrated in Fig. 3, these

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combined features are processed through a depth-separable convolution to generate the final refined edge maps, R_e and D_e. By integrating traditional geometric priors with comprehensive deep semantic features, the ECM ensures precise and robust boundary representations.

Taking the RGB stream as an example to illustrate the workflow of the ECM. After the RGB stream features R_i, ($i \in [1, 5]$) from the backbone network enter the ECM, they first undergo receptive field activation through a 5×5 convolutional layer and a ReLU layer. Subsequently, from R_5 to R_1, a $2\times$ upsampling is performed layer by layer, followed by a 3×3 convolution, and the semantic information from various levels is fused through element-wise addition. Let $f(R) = Activation(R)$ and $g(X) = Upsample(X)$. We define the intermediate results H_k iteratively, starting from $k = 5$:

$$\begin{aligned} H_5 &= g(f(R_5)) \\ H_4 &= g(f(R_4) \oplus H_5) \\ H_3 &= g(f(R_3) \oplus H_4) \\ H_2 &= g(f(R_2) \oplus H_3) \end{aligned} \tag{1}$$

The final result E_{rgb} is then computed as:

$$E_{rgb} = g(f(R_1) \oplus H_2) \tag{2}$$

where *Activation* represents a 5×5 convolutional layer with grouped convolution, $Upsample$ denotes a $2\times$ upsampling, and $\oplus$ indicates element-wise addition. This effectively combines semantics from various levels to construct more precise edge features.

Finally, to leverage the structural guidance of the NAMLab, the initial edge features E_{rgb} and E_{depth} (96 channels) derived from the cascaded sampling are concatenated with the 1-channel NAMLab edge maps (RGB_{NAMLab} and $depth_{NAMLab}$) at the module's output. By integrating these traditional geometric priors with deep semantic features, the combined maps are then processed through a depth-separable convolution to generate the final refined edge maps R_e and D_e, as illustrated in Fig. 3 and defined in formula (3). This hybrid mechanism ensures that the ECM produces more precise and robust boundary representations by effectively combining handcrafted structural cues with learned multi-level semantics.

$$R_e = Conv\big(Concat(E_{rgb},\ RGB_{NAMLab})\big) \tag{3}$$

where E_{rgb} represents the initial edge features output by the ECM module, RGB_{NAMLab} denotes the edge map extracted from the RGB image using the NAMLab algorithm, and *Conv* stands for depth-separable convolution.

C. Cross-Attention Fusion Module (CAFM)

The design rationale of the Cross-Attention Fusion Module (CAFM) is to achieve a balance between fusion efficiency and representation capability. By introducing local window proposals, the CAFM aims to perform efficient cross-modal interaction that reduces the complexity of traditional global attention to a manageable level. This design ensures thorough feature exchange between modalities while maintaining a low computational footprint.

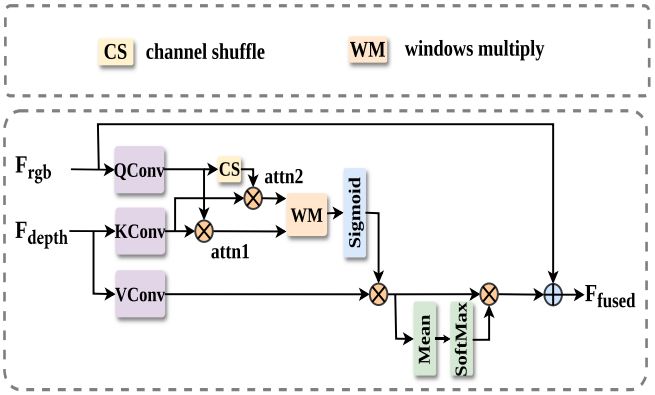


Fig. 4. Illustration of the internal structure of our Cross-Attention Fusion Module (CAFM). 'QConv', 'KConv', 'VConv' denote 5×5 convolutional layers generating *Query*, *Key*, and *Value* in (4), respectively. 'attn1' represents attention weights from element-wise multiplication of *Query* and *Key*. 'attn2' represents attention weights from element-wise multiplication after channel-shuffling *Query* and *Key*.

In traditional cross-attention implementations, the image tensors often need to be serialized. For example, ViT [40] first divided a 224×224 image into a sequence of 16×16 patches (total 196 tokens). However, this approach typically faces the following challenges: (i) The computational complexity of the attention matrix calculation is $O(n^2)$ in long sequence scenarios; (ii) The robustness to input distribution changes still needs improvement; (iii) forcing each token to establish connections with all other tokens globally often leads to computational redundancy.

By addressing these issues, and inspired by the Cross-Attention mechanism in Transformer, we have proposed a cross-attention mechanism based on the convolution and local window proposals. First, the input tensor undergoes receptive field activation through a 5×5 convolutional layer, and in order to achieve a balance between the computational cost and the performance, a grouped convolution is employed to implement channel decoupling, as shown in following formulas presented in (4):

$$\begin{aligned} Query &= Conv_{5\times5}(F_{rgb}) \\ Key &= Conv_{5\times5}(F_{depth}) \\ Value &= Conv_{5\times5}(F_{depth}) \end{aligned} \tag{4}$$

where F_{rgb} and F_{depth} represent the features from the RGB modality and the depth modality, respectively. By using the RGB features as *Query* and the depth features as *Key* and *Value*, a feature guidance is implemented in an asymmetric form, enabling the RGB features to actively "inquire about" geometric information in the depth features, thereby allowing the depth discontinuities at the object edges to enhance the RGB boundaries, while the depth consistency in the planar regions can suppress the RGB texture noise.

Unlike the traditional methods that use dot product attention to calculate cosine similarity for attention weighting, in this paper, we have proposed a local window proposal-based method to reduce computational cost, as shown in Fig. 4. The specific process is as follows: First, *Query* and *Key* tensors are appropriately fused through element-wise multiplication to

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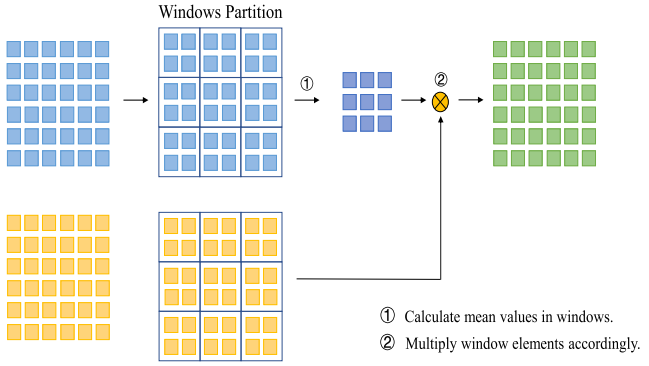


Fig. 5. Illustration of our local window proposal-based cross-attention fusion module.

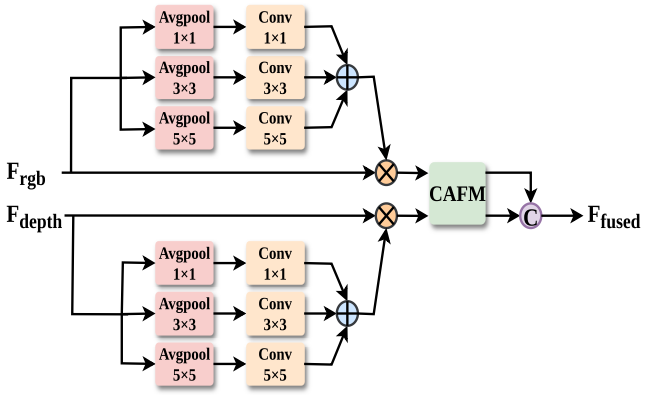


Fig. 6. Illustration of our proposed Cross-Modal multi-scale Attention-weighting Module (CMAM).

obtain initial attention weights (termed as *attn*1). This fusion is typically coarse and requires further processing. Since we had extensively used grouped convolution methods previously, to ensure thorough cross-channel information exchange, we have calculated a second *attn*2 after randomly shuffling the channels of *Query* and *Key*. Subsequently, we have divided both *attn*1, *attn*2 tensors into windows, as illustrated in Fig. 5. In practical implementation, we divide the tensor into numerous 4×4 windows. Then, *attn*1 calculates the mean within each independent window, resulting in a tensor with dimensions 1/4 of the original. We refer to each element of this tensor as a local window proposal. The elements of this tensor represent commonalities between the two modalities, enabling mutual enhancement and suppression of features between modalities. Finally, each element of this tensor is expanded back to the original window size through a broadcasting mechanism and multiplied by the corresponding elements within the window of the *attn*2, thereby obtaining attention weights (denoted as *Attention*) that have undergone modal commonality suppression or enhancement through local window proposals, which is presented in the following formula:

$$Attention = WindowPartition(attn2) \otimes BC\Big(Mean\big(WP(attn1)\big)\Big) \tag{5}$$

where $attn1 = Query \otimes Key$, $attn2 = ChannelShuffle(Query) \otimes ChannelShuffle(Key)$. *BC* represents the broadcast operation, and $\otimes$ signifies element-wise multiplication.

Simultaneously, employing channel attention mechanisms for gating can effectively integrate the semantic information across multiple channels, as shown in the following formula (6):

$$Attention' = Attention \otimes CA(Attention) \tag{6}$$

Finally, the integrity of the original RGB features is preserved through residual connections, preventing gradient vanishing and allowing the network to autonomously regulate fusion intensity, automatically attenuating fusion weights when depth features contain significant noise. This can be expressed as in the following formula:

$$CAFM(F_{rgb}, F_{depth}) = F_{rgb} + F_{fused} \tag{7}$$

where $F_{fused} = Attention' \otimes Value$.

The computational complexity of this method is $O(nc)$, with a parameter count of $O(3k^2c)$, where $k = 5$ is the convolution kernel size and c is the number of channels. In contrast, the standard cross-attention implementation has a computational complexity of $O(n^2c)$ and a parameter count of $O(3c^2)$. When the channel number is 256, the computational cost is reduced by approximately 90%, while still maintaining satisfactory performance.

D. Cross-Modal Multi-Scale Attention-Weighting Module (CMAM)

The goal of the Cross-Modal Multi-scale Attention-weighting Module (CMAM) is to dynamically evaluate the reliability of RGB and depth streams across multiple scales. By adaptively adjusting their contributions, the CMAM effectively suppresses unreliable depth cues in noisy environments and enhances the robustness of the network against challenging backgrounds.

When processing the fusion of the RGB and depth modalities, these two modalities often have significant semantic disparities. The RGB modality typically contains richer edge details, but it may include texture noise. The depth modality contains more foreground-background information, but it loses critical textures and details. However, controlling fusion weights presents a major challenge. We combine the spatial pyramid pooling with cross-attention, achieving breakthroughs in the following three dimensions: (i) In the spatial dimension, we have implemented progressive fusion from local to global features. (ii) In the channel dimension, we have used depth-wise convolution to achieve modal decoupling. (iii) In the cross dimension, we have used bidirectional attention to enhance semantic alignment.

The module architecture is shown in Fig. 6. The specific workflow is as follows (by using the RGB stream as an example; the depth stream follows the same principle). First, the RGB stream feature $F_r \in \mathbb{R}^{H \times W \times C}$ enters the module and passes through 5×5, 3×3, and 1×1 adaptive average pooling layers with the corresponding convolutional layers, thereby obtaining receptive fields for the local details, medium-range

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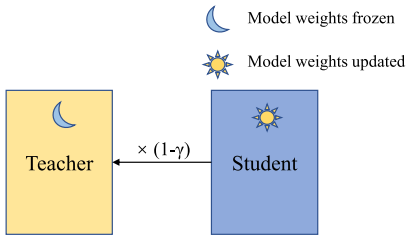


Fig. 7. Illustration of how knowledge distillation in our refinement stage work.γ is the momentum coefficient for model updating.

semantics, and global estimation respectively as presented in the following formula:

$$F_{ri} = Conv_{jxj}\big(avgpool_{jxj}\ (F_r)\big) \tag{8}$$

where $i \in \{1,\ 2,\ 3\}$, $j \in \{1,\ 3,\ 5\}$.

Then, a dynamic weight learning mechanism is employed to fuse the above-obtained weight values in a learnable manner as shown in the following formula:

$$F_{rgb} = \alpha \times F_{r1} + \beta \times F_{r2} + \gamma \times F_{r3} \tag{9}$$

where α, β, γ are randomly initialized learnable parameters.

Finally, since the CAFM itself is asymmetric, a bidirectional cross-modal attention fusion is performed through the CAFM module, followed by concatenation in the channel dimension, establishing bidirectional information exchange between the RGB and the depth modalities as explained in the following formula:

$$\begin{aligned} F_{fused} = Concat\big(&F_{rgb} \otimes CAFM(F_r, F_d),\\ &F_{depth} \otimes CAFM(F_d, F_r)\big) \end{aligned} \tag{10}$$

where F_{fused} represents the fused features, $\otimes$ signifies element-wise multiplication. F_{rgb} and F_{depth} are the modal weight values obtained by the aforementioned methods.

E. Two-Stage Refinement

Our training method consists of two stages. The first stage employs normal training (termed as the general stage), whereas the second stage (termed as the refinement stage), as shown in Fig. 7, uses the model parameters obtained from the general stage to regenerate the pseudo-labels required for training, replacing the initial scribble annotations. The refinement stage adopts the model from the general stage as a teacher model and employs knowledge distillation methods for model refinement and fine-tuning. Specifically, the teacher model generated from the general stage is first loaded, with training conducted only on the student model. After each round of backpropagation, the student model's parameters are updated to the teacher model using a certain momentum coefficient. Finally, the teacher model fine-tuned during the refinement stage is used as the final model for prediction. The model update method can be expressed as follows:

$$\Psi'_{teacher} = \gamma \times \Psi_{teacher} + (1-\gamma) \times \Psi_{student} \tag{11}$$

where Ψ represents all the model parameters, $\Psi_{teacher}$ represents the teacher model parameters before updating, $\Psi'_{teacher}$ represents the teacher model parameters after updating, $\Psi_{student}$ represents the student model parameters, and γ represents the momentum coefficient for model updating.

F. Loss Function

As we have mentioned before, our training is divided into two stages: the general stage and the refinement stage. The loss functions for the two stages differ slightly.

1) General Stage: In this paper, we have adopted the Pixel Position Aware loss function (PPA) proposed by F3Net [41] to enhance the model's attention to the boundary regions of the salient objects. To make the model focuses more on the segmentation quality of the salient object boundary regions, a weight α is assigned to each pixel. The parameter α is calculated based on the difference between the central pixel and its surrounding environment, as shown in the following equation (12).

$$\alpha^s_{ij} = \left| \frac{\sum_{m,n \in A_{ij}} gt^s_{mn}}{\sum_{m,n \in A_{ij}} 1} - gt^s_{ij} \right| \tag{12}$$

where A_{ij} represents the area surrounding the pixel (i, j), and for all the pixels, $\alpha^s_{ij} \in [0, 1]$.

As shown in following formulas (13) and (14), the improved loss function assigns a weight to each pixel based on the difference between the pixel and its surrounding environment, with higher weights typically assigned to watershed regions between the salient objects and the background, where the segmentation quality is crucial for SOD results.

$$L^s_{wbce} = -\frac{\sum_{i=1}^{H}\sum_{j=1}^{W}(1+\gamma\alpha_{ij})\sum_{l=0}^{1}\mathbf{1}(g^s_{ij}=l)log\mathbf{Pr}(p^s_{ij}=l|\Psi)}{\sum_{i=1}^{H}\sum_{j=1}^{W}\gamma\alpha_{ij}} \tag{13}$$

$$L^s_{wiou} = 1 - \frac{\sum_{i=1}^{H}\sum_{j=1}^{W}(gt^s_{ij}\cdot p^s_{ij})\cdot(1+\gamma\alpha^s_{ij})}{\sum_{i=1}^{H}\sum_{j=1}^{W}(gt^s_{ij}+p^s_{ij}-gt^s_{ij}\cdot p^s_{ij})\cdot(1+\gamma\alpha^s_{ij})} \tag{14}$$

where $\mathbf{1}(\cdot)$ is an indicator function, and γ is a hyperparameter, the representation function $l \in \{0, 1\}$ denotes two types of labels, the g^s_{ij} and p^s_{ij} represent the ground truth value and the prediction result of the pixel at the position (i, j) in the image, respectively. Ψ represents all the parameters of the model, and $\mathbf{Pr}(p^s_{ij} = l|\Psi)$ denotes the predicted probability.

Compared to the BCE, the L^s_{wbce} focuses more on the hard pixels. Furthermore, the local structural information has been encoded as L^s_{wbce}, helping the model focus on a larger receptive field rather than the individual pixels.

Since edge pixels typically constitute only a minimal portion of the total image pixels, the cross-entropy loss tends to bias toward the background due to the pixel-level cumulative calculation, making it difficult to effectively learn the edge features. By calculating the overlap (Dice coefficient [42]) between the predicted and the actual edge regions, the Dice loss [43], [44] can directly ignore a large number of background pixels and focus only on the global matching of the

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edge regions, thus alleviating the sample imbalance problem. The Dice Loss dynamically adjusts the loss weights through the ratio of the intersection to the total pixels, making the model more attentive to minority-class regions such as the edges. The supervising boundary regions with Dice Loss can incorporate explicit geometric prior knowledge (such as the gradient features of edges) into the feature learning process, enhancing the model's sensitivity to the edge structures as shown in the following equation:

$$L^s_{dice} = \frac{2 \cdot \sum_i^N p_i \cdot g_i}{\sum_i^N p_i^2 + \sum_i^N g_i^2} \tag{15}$$

where the summation range is over N pixels, and p_i, g_i represent the predicted value and the ground truth value, respectively.

Then, losses are computed separately for the edge map resulting from the RGB-D fusion (L^s_{fused}) and the edge map predicted from the depth stream (L^s_{depth}). They integrate the local structural information, generate different weights for all pixels, and introduce pixel constraints (L^s_{wbce}) and global constraints (L^s_{wiou}), which can better guide network learning and produce clear details:

$$\begin{aligned} L^s_{fused} &= L^s_{wbce}(GT_{edge}, E_{fused}) \\ &\quad + L^s_{wiou}(GT_{edge}, E_{fused}) \\ &\quad + L^s_{dice}(GT_{NAMLab}, E_{fused}) \end{aligned} \tag{16}$$

$$\begin{aligned} L^s_{depth} &= L^s_{wbce}(GT_{edge}, E_{depth}) \\ &\quad + L^s_{wiou}(GT_{edge}, E_{depth}) \\ &\quad + L^s_{dice}(GT_{NAMLab}, E_{depth}) \end{aligned} \tag{17}$$

Based on the above discussion, our proposed boundary loss function is shown in the following formula (18).

$$L^s_{edge} = L^s_{fused} + L^s_{depth} \tag{18}$$

In both the general stage and the refinement stage, our supervision of saliency maps is divided into two parts. The first part is the supervision of the initial coarse prediction map, represented as S_1 in Fig. 2, and the second part is the refined prediction map after secondary fusion with edge features, represented as S_2 in Fig. 2.

For the first part of the coarse prediction map, we have the following objective function:

$$L^c_{pce} = -\frac{1}{|S|}\sum_{p\in S}(y_p \log \hat{y}^c_p + (1-y_p)\log(1-\hat{y}^c_p)) \tag{19}$$

For the second part of the refinement prediction map, we have the following objective function:

$$L^r_{pce} = -\frac{1}{|S|}\sum_{p\in S}(y_p \log \hat{y}^r_p + (1-y_p)\log(1-\hat{y}^r_p)) \tag{20}$$

where S represents all pixels annotated with scribbles, $\hat{y}^c_p$ denotes the coarse saliency prediction map without further fusion with edge features, and $\hat{y}^r_p$ is the prediction map after edge fusion and refinement.

Therefore, our proposed complete loss function is defined as shown, in the following equation (21):

$$L^s = \lambda_{edge} L^s_{edge} + L^c_{pce} + L^r_{pce} + \lambda_{sm}(L^c_{sm} + L^r_{sm}) \tag{21}$$

where L^c_{sm} and L^r_{sm} represent the structure-aware smoothness loss terms for $\hat{y}^c_p$ and $\hat{y}^r_p$. λ_{sm} and λ_{edge} are the weights for the smoothness term and the edge loss, respectively.

2) Refinement Stage: The complete loss function for our proposed refinement stage is as shown, in the following equation (22):

$$\begin{aligned} L^s &= \lambda_{edge} L^s_{edge} + L^s_{wbce}(GT, \hat{y}^c) + L^s_{wiou}(GT, \hat{y}^c) \\ &\quad + L^s_{wbce}(GT, \hat{y}^r_p) + L^s_{wiou}(GT, \hat{y}^r_p) \\ &\quad + \lambda_{sm}(L^c_{sm} + L^r_{sm}) \end{aligned} \tag{22}$$

where λ_{sm} and λ_{edge} are the weights for the smoothness term and the edge loss. L^c_{sm} and L^r_{sm} represent the structure-aware smoothness loss terms for $\hat{y}^c_p$ and $\hat{y}^r_p$.

IV. Experimental Results

A. Evaluation Metrics

To evaluate the performance of the proposed NESS-Net, we employ four widely used metrics: Structure-measure (S_m) [45], adaptive Enhanced-alignment measure (E^{adp}_m) [46], adaptive F-measure (F^{adp}_β) [47], and Mean Absolute Error (MAE) [48]. Following the standard evaluation protocol in RGB-D SOD, we adopt the adaptive thresholding strategy [48] for both E_m and F_β to provide a comprehensive assessment of the prediction quality.

B. Datasets and Implementation Details

To ensure fair comparison with the existing RGB-D saliency detection models, and by following the work done in [9], [10], and [49], we have adhered to the conventional training setup, where the training set consists of 1,485 images from the NJU2K dataset [2] and 700 images from the NLPR dataset [3]. Then, we have tested the performance of our model and the competing models on 9 test sets: NJU2K, NLPR, LFSD [50], DES [51], SSD [52], SIP [26], DUTLF-Depth [1], ReDWeb-S [53], and STERE [54]. Our model is trained on a server equipped with 2 NVIDIA RTX 3090 GPUs. To ensure consistency in the input size, we have resized the input images to a resolution of 384 × 384 and applied data augmentation operations such as random flipping, rotation, and cropping to the training images. We have trained the NESS-Net using the Adam optimizer with a batch size of 4 per GPU. The initial learning rate is set to $5e-5$, decaying to 0.9 times the original rate every 20 epochs. We have set λ_{sm} to 0.1, λ_{edge} to 0.1, and the momentum coefficient (γ) for model updates to 0.999.

Our model is implemented using PyTorch. In the first training stage (i.e., the general stage), we have set the maximum number of epochs to 35. In the second stage (i.e., the refinement stage), the maximum number of epochs is set to 30. The refinement stage uses the model parameters from the general stage to obtain pseudo-labels for the training set, which, after post-processing with denseCRF [55], replace the scribble labels as the new GT maps. The first stage training takes 5.5 hours, and the second stage takes 4 hours.

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TABLE I
COMPARATIVE EXPERIMENTAL RESULTS WITH STATE-OF-THE-ART FULLY SUPERVISED MODELS AND SCRIBBLE-SUPERVISED MODELS, INCLUDING S-MEASURE ($S_m \uparrow$), ADAPTIVE E-MEASURE ($E_m^{adp} \uparrow$), ADAPTIVE F-MEASURE ($F_\beta^{adp} \uparrow$), MEAN ABSOLUTE ERROR METRICS ($MAE \downarrow$), FLOPS (G) $\downarrow$, AND PARAMETERS (M) $\downarrow$. THE BEST RESULTS ARE HIGHLIGHTED IN RED FONT, AND THE SECOND-BEST RESULTS ARE HIGHLIGHTED IN BLUE FONT. $\uparrow$ AND $\downarrow$ REPRESENT THAT HIGHER VALUES AND LOWER VALUES ARE BETTER FOR THE CORRESPONDING METRICS, RESPECTIVELY. "OURS[1] (NESSNET-G)" IN THE TABLE INDICATES THE MODEL PERFORMANCE AT THE GENERAL STAGE, WHEREAS "OURS[2] (NESSNET)" INDICATES THE MODEL PERFORMANCE AT THE REFINEMENT STAGE

Year	Method	Overall		NJU2K				NLPR				LFSD				SIP				SSD				ReDWeb-S				DUTLF-Depth				STERE				DES			
		FLOPs (G) $\downarrow$	Params (M) $\downarrow$	$S_m \uparrow$	$F_\beta^{adp} \uparrow$	$E_m^{adp} \uparrow$	$MAE \downarrow$	$S_m \uparrow$	$F_\beta^{adp} \uparrow$	$E_m^{adp} \uparrow$	$MAE \downarrow$	$S_m \uparrow$	$F_\beta^{adp} \uparrow$	$E_m^{adp} \uparrow$	$MAE \downarrow$	$S_m \uparrow$	$F_\beta^{adp} \uparrow$	$E_m^{adp} \uparrow$	$MAE \downarrow$	$S_m \uparrow$	$F_\beta^{adp} \uparrow$	$E_m^{adp} \uparrow$	$MAE \downarrow$	$S_m \uparrow$	$F_\beta^{adp} \uparrow$	$E_m^{adp} \uparrow$	$MAE \downarrow$	$S_m \uparrow$	$F_\beta^{adp} \uparrow$	$E_m^{adp} \uparrow$	$MAE \downarrow$	$S_m \uparrow$	$F_\beta^{adp} \uparrow$	$E_m^{adp} \uparrow$	$MAE \downarrow$	$S_m \uparrow$	$F_\beta^{adp} \uparrow$	$E_m^{adp} \uparrow$	$MAE \downarrow$
Fully-Supervised Models																																							
ICCV 2021	VST	31.1	83.1	0.900	0.828	0.899	0.046	0.918	0.861	0.945	0.026	0.847	0.796	0.869	0.079	0.872	0.850	0.917	0.057	0.889	0.817	0.906	0.045	0.759	0.698	0.783	0.112	0.943	0.948	0.969	0.024	0.902	0.830	0.910	0.042	0.916	0.876	0.961	0.025
AAAI 2021	RD3D	50.9	46.9	0.924	0.882	0.927	0.034	0.925	0.877	0.951	0.025	0.868	0.850	0.895	0.069	0.896	0.880	0.926	0.044	0.803	0.733	0.860	0.082	0.628	0.576	0.703	0.176	0.932	0.939	0.960	0.031	0.914	0.861	0.923	0.038	0.937	0.912	0.973	0.019
TPAMI 2022	UCNet	16.2	27.0	0.894	0.899	0.937	0.044	0.918	0.904	0.957	0.026	0.858	0.859	0.898	0.072	0.876	0.882	0.925	0.049	0.865	0.854	0.907	0.049	0.713	0.710	0.794	0.130	0.871	0.866	0.910	0.059	0.900	0.898	0.944	0.039	0.925	0.916	0.965	0.021
CVPR 2022	PASNet	85.1	219.9	0.906	0.905	0.946	0.042	0.916	0.903	0.956	0.029	0.855	0.852	0.891	0.079	0.880	0.877	0.926	0.052	0.874	0.851	0.914	0.046	0.722	0.711	0.799	0.126	0.875	0.875	0.920	0.064	0.900	0.893	0.942	0.043	0.930	0.924	0.971	0.022
PR 2023	HINet	49.0	98.9	0.914	0.896	0.919	0.037	0.906	0.847	0.933	0.031	0.851	0.846	0.885	0.077	0.886	0.878	0.921	0.048	0.832	0.789	0.888	0.068	0.668	0.644	0.726	0.167	0.889	0.883	0.922	0.053	0.905	0.870	0.919	0.041	0.944	0.904	0.974	0.018
TIP 2023	LSNet	1.2	4.6	0.911	0.895	0.921	0.039	0.918	0.879	0.954	0.025	0.833	0.851	0.873	0.085	0.886	0.881	0.926	0.050	0.763	0.712	0.858	0.090	0.600	0.566	0.691	0.192	-	-	-	-	0.871	0.850	0.912	0.055	0.926	0.906	0.969	0.021
TOMM 2024	HFILNet	144.6	241.9	0.936	0.933	0.939	0.025	0.943	0.922	0.970	0.016	0.885	0.878	0.910	0.058	0.909	0.916	0.942	0.034	0.904	0.891	0.934	0.032	0.760	0.762	0.783	0.109	0.950	0.948	0.968	0.019	0.922	0.903	0.934	0.029	0.945	0.937	0.980	0.014
KBS 2024	SPCNet	275.8	154.4	0.921	0.893	0.931	0.033	0.931	0.904	0.962	0.020	0.876	0.858	0.899	0.062	0.888	0.883	0.928	0.044	0.886	0.860	0.918	0.042	0.731	0.694	0.765	0.124	0.895	0.882	0.925	0.047	0.903	0.863	0.926	0.038	0.941	0.933	0.981	0.016
TNNLS 2024	RD3D+	43.5	28.9	0.927	0.895	0.919	0.033	0.933	0.884	0.958	0.022	0.861	0.842	0.873	0.076	0.891	0.880	0.923	0.046	0.837	0.794	0.896	0.060	0.663	0.623	0.724	0.165	0.936	0.945	0.964	0.030	0.913	0.874	0.921	0.039	0.949	0.915	0.978	0.017
TPAMI 2024	VST++	32.7	72.1	0.917	0.855	0.913	0.035	0.930	0.894	0.957	0.022	0.876	0.829	0.883	0.065	0.890	0.874	0.930	0.045	0.894	0.844	0.917	0.039	0.762	0.695	0.775	0.109	0.953	0.959	0.974	0.019	0.915	0.857	0.922	0.036	0.931	0.916	0.975	0.019
TCSVT 2025	HENet	10.8	10.4	0.934	0.924	0.926	0.027	0.942	0.918	0.971	0.016	0.900	0.891	0.906	0.049	0.914	0.918	0.946	0.032	0.892	0.860	0.915	0.036	0.750	0.750	0.775	0.117	0.949	0.942	0.966	0.019	0.926	0.914	0.939	0.029	0.955	0.940	0.986	0.013
ESWA 2025	PLFRNet	25.1	41.9	0.931	0.928	0.934	0.026	0.941	0.927	0.971	0.016	0.891	0.885	0.911	0.050	0.911	0.919	0.941	0.034	0.887	0.864	0.917	0.038	0.757	0.768	0.785	0.111	0.951	0.951	0.968	0.018	0.924	0.909	0.937	0.028	0.942	0.936	0.979	0.015
ESWA 2025	ACINet	39.1	85.7	0.932	0.926	0.926	0.026	0.939	0.922	0.970	0.016	0.889	0.881	0.913	0.050	0.908	0.914	0.939	0.034	0.885	0.853	0.913	0.042	0.765	0.769	0.785	0.106	0.947	0.944	0.966	0.020	0.921	0.903	0.933	0.030	0.945	0.936	0.981	0.014
Scribble-Supervised Models																																							
TIP 2022	DENet	179.7	18.6	0.874	0.861	0.899	0.053	0.898	0.858	0.943	0.031	0.822	0.821	0.865	0.087	0.846	0.825	0.904	0.062	0.823	0.778	0.859	0.073	0.704	0.701	0.763	0.138	0.855	0.835	0.901	0.064	0.874	0.845	0.908	0.051	0.903	0.886	0.959	0.027
TPAMI 2024	DSP	83.7	130.8	0.886	0.843	0.897	0.056	0.897	0.828	0.932	0.035	0.831	0.838	0.875	0.096	0.875	0.847	0.912	0.060	0.859	0.798	0.885	0.060	0.700	0.691	0.768	0.142	0.876	0.839	0.908	0.066	0.879	0.839	0.910	0.059	0.913	0.877	0.963	0.028
TCSVT 2024	MIRV-F	245.3	114.2	0.890	0.878	0.901	0.046	0.905	0.875	0.946	0.028	0.828	0.824	0.857	0.085	0.865	0.846	0.907	0.057	0.838	0.791	0.877	0.064	0.715	0.715	0.763	0.131	0.715	0.870	0.763	0.051	0.891	0.870	0.917	0.043	0.917	0.899	0.963	0.024
TCSVT 2024	MIRV	37.1	63.6	0.889	0.882	0.907	0.045	0.914	0.891	0.954	0.025	0.849	0.852	0.886	0.072	0.876	0.963	0.923	0.049	0.855	0.828	0.897	0.054	0.704	0.708	0.757	0.131	0.877	0.870	0.910	0.053	0.890	0.874	0.918	0.041	0.928	0.922	0.971	0.019
TGRS 2025	LightEMNet	27.4	4.8	0.861	0.862	0.893	0.054	0.881	0.858	0.936	0.033	0.794	0.800	0.851	0.095	0.814	0.804	0.885	0.076	0.816	0.783	0.883	0.067	0.640	0.625	0.703	0.158	0.873	0.871	0.927	0.048	0.870	0.864	0.907	0.050	0.867	0.857	0.934	0.037
	Ours[2] (NESSNet-G)	143.5	183.6	0.888	0.894	0.920	0.039	0.904	0.892	0.957	0.024	0.855	0.864	0.888	0.065	0.862	0.874	0.926	0.048	0.842	0.829	0.899	0.051	0.716	0.730	0.764	0.116	0.894	0.901	0.937	0.041	0.878	0.874	0.922	0.043	0.905	0.908	0.962	0.022
	Ours[2] (NESSNet)	143.5	183.6	0.897	0.904	0.924	0.035	0.908	0.896	0.959	0.023	0.867	0.874	0.897	0.058	0.871	0.884	0.932	0.044	0.855	0.849	0.912	0.045	0.727	0.737	0.766	0.113	0.910	0.920	0.951	0.033	0.881	0.875	0.923	0.041	0.923	0.922	0.975	0.019

C. Comparison and Evaluation With SOTA Models

1) Quantitative Comparison: We have compared our NESS-Net model with the state-of-the-art fully supervised models and scribble-supervised models, as shown in Table I. The fully supervised models used for the comparison include: UCNet [58], VST [59], VST++ [60], PASNet [61], HINet [50], RD3D [62], RD3D+ [63], LSNet [64], HFILNet [65], SPCNet [66], HENet [67], PLFRNet [68] and ACINet [69], as well as scribble-supervised models: DENet [9], DSP [49], MIRV [10] and LightEMNet [70]. Unlike other scribble-supervised models primarily designed for natural scene images, LightEMNet is specifically developed for the domain of Optical Remote Sensing Image Salient Object Detection (ORSI-SOD).

It can be seen that in the field of scribble supervision, our model achieves the best performance among the SOTA models on the majority of metrics across 9 datasets. Furthermore, on certain datasets such as LFSD [50], SIP [26] and NLPR [3], the performance of our model surpasses most fully supervised models on most metrics, and it ranks among the top performers of the fully supervised models on most datasets.

2) Qualitative Comparison: As shown in Fig. 8, we have provided a visual comparison between the NESS-Net model and other models. It can be seen that our model's predictions are generally more accurate and cleaner compared to the predictions of other models. From Fig. 8, we can see that our model's visualization results outperform all scribble-supervised models as well as most fully-supervised models. This indicates that our model exhibits better robustness to noise and can learn more accurate foreground details, easily distinguishing between the foreground and background of the images.

D. Ablation Studies

In order to investigate the role of different modules in the NESS-Net, we have proposed a series of comparative experiments.

1) Backbone: We have adopted the Swin-B as the backbone network. To verify the importance of using the Swin-Transformer as the backbone network, we have validated its effectiveness by replacing the Swin-Transformer with the ResNet50 [11] and VGG16 [12] as backbone networks.

As shown in Table II, it can be observed that the Swin-Transformer demonstrated its superior performance across the seven datasets provided, which indicates that using the Swin-Transformer as the backbone network is highly effective.

2) CAFM and CMAM: To verify the effectiveness of the Cross-Attention Fusion Module (CAFM) and the Cross-Modal multi-scale Attention-weighting Module (CMAM), we conducted ablation experiments targeting these components.

As shown in Table III, removing either the CAFM or CMAM leads to varying degrees of performance degradation, confirming their positive contribution to the final model.

a) Comparison with Standard Cross-Attention: Furthermore, to substantiate the efficiency of our proposed CAFM, we compared it against the standard global Cross-Attention

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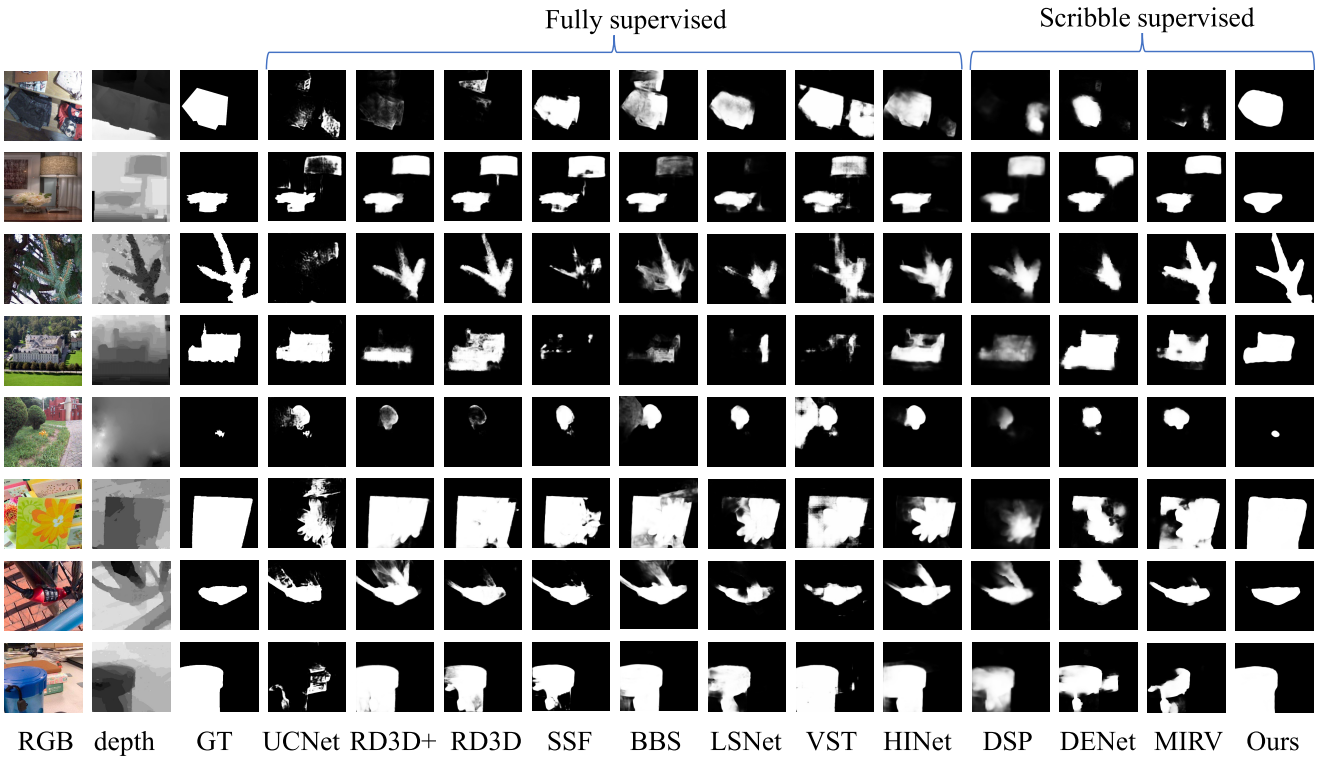


Fig. 8. Qualitative comparison between the NESS-Net model and SOTA models, including fully supervised models: SSF [56], BBS [57], UCNet [58], VST [59], HINet [50], RD3D [62], RD3D+ [63], LSNet [64], as well as scribble-supervised models: DENet [9], DSP [49], and MIRV [10]. 'Ours' represents the results of our proposed refinement stage.

mechanism in terms of computational cost and resource consumption.

The experiments were conducted with an input feature size of $256 \times 48 \times 48$.

As detailed in Table IV, the CAFM significantly reduces the computational complexity (FLOPs) by approximately 90% (from 0.469G to 0.044G) and nearly halves the GPU memory usage compared to the standard implementation.

Although the standard attention typically benefits from highly optimized matrix multiplication libraries (e.g., cuBLAS), resulting in a slightly lower latency (0.25 ms vs. 0.36 ms), the massive reduction in FLOPs and parameters makes the CAFM much more suitable for resource-constrained scenarios while maintaining competitive performance.

b) Influence Of Window Size in the CAFM: The window size $w \times w$ in the Cross-Attention Fusion Module (CAFM) determines the granularity of the local window proposals, balancing the trade-off between receptive field and detail preservation.

To analyze its impact, we conducted experiments with three window settings: 2×2, 4×4 (ours), and 6×6. The quantitative results are summarized in Table V.

When the window size is small (2×2), although the computational cost is minimized, the receptive field is too limited to capture sufficient contextual information, leading to fragmented proposals and a drop in overall performance compared to the 4×4 setting (e.g., S_m drops from 0.922 to 0.916 on DES dataset).

Conversely, increasing the window size to 6×6 does not yield further improvements; instead, it causes performance degradation.

We hypothesize that larger windows excessively smooth out local features, merging distinct fine details into coarse representations, which is detrimental for precise salient object segmentation.

Thus, the 4×4 configuration is adopted as the optimal choice, achieving the best balance between accuracy and structural integrity across all datasets.

c) Diagnostic Measure for Modality Re-Calibration: Beyond standard performance metrics, we further investigate the internal behavior of the CMAM to understand how it handles multi-modal inputs, particularly under challenging conditions.

Since the mechanism (10) operates implicitly via element-wise multiplication, observing raw attention weights alone is insufficient to quantify the actual impact on feature representation. Therefore, instead of using standard performance metrics, we propose to trace the "energy flow" of features through the module.

Intuitively, if the CMAM identifies a modality as noisy, the total activation magnitude (energy) of that modality's features should decrease relative to the other modality after processing. To formalize this, we define a diagnostic measure ΔR (local depth energy ratio shift).

To formalize this, let $x^{\text{rgb}}, x^{\text{dep}} \in \mathbb{R}^{C\times H\times W}$ denote the input features to the gating mechanism, and $\tilde{x}^{\text{rgb}}, \tilde{x}^{\text{dep}}$ be the output

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TABLE II

ABLATION STUDY USING SWIN-B, RESNET50, AND VGG16 AS MODEL BACKBONES. ROWS REPRESENT MODELS, COLUMNS ARE GROUPED BY DATASET THEN METRIC. THE BEST RESULTS FOR EACH METRIC ON EACH DATASET ARE HIGHLIGHTED IN RED

Backbone	Params ↓ (M)	FLOPs ↓ (G)	DES				SIP				LFSD				NJU2K				NLPR				STERE				DUTLF-Depth			
			$S_m\uparrow$	$F_\beta^{\text{adp}}\uparrow$	$E_m^{\text{adp}}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{\text{adp}}\uparrow$	$E_m^{\text{adp}}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{\text{adp}}\uparrow$	$E_m^{\text{adp}}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{\text{adp}}\uparrow$	$E_m^{\text{adp}}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{\text{adp}}\uparrow$	$E_m^{\text{adp}}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{\text{adp}}\uparrow$	$E_m^{\text{adp}}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{\text{adp}}\uparrow$	$E_m^{\text{adp}}\uparrow$	$MAE\downarrow$
Swin-B	86.68	44.66	0.922	0.918	0.973	0.019	0.876	0.885	0.935	0.043	0.871	0.873	0.897	0.058	0.900	0.901	0.921	0.036	0.908	0.892	0.957	0.024	0.884	0.873	0.922	0.042	0.911	0.917	0.950	0.034
ResNet50	25.56	12.14	0.908	0.885	0.951	0.026	0.854	0.834	0.912	0.057	0.832	0.824	0.874	0.086	0.875	0.857	0.890	0.052	0.891	0.850	0.937	0.033	0.872	0.851	0.910	0.051	0.858	0.836	0.897	0.065
VGG16	138.36	45.22	0.854	0.879	0.939	0.030	0.853	0.844	0.912	0.058	0.819	0.823	0.871	0.090	0.872	0.863	0.899	0.052	0.892	0.870	0.944	0.031	0.868	0.855	0.916	0.052	0.852	0.843	0.898	0.065

TABLE III

ABLATION STUDY RESULTS OF THE CAFM AND CMAM MODULES, WHERE "✓" DENOTES THAT THE CORRESPONDING MODULE IS USED. THE BEST RESULTS ARE HIGHLIGHTED IN RED

CAFM	CMAM	DES				SIP				LFSD				NJU2K				NLPR			
		$S_m\uparrow$	$F_\beta^{\text{adp}}\uparrow$	$E_m^{\text{adp}}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{\text{adp}}\uparrow$	$E_m^{\text{adp}}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{\text{adp}}\uparrow$	$E_m^{\text{adp}}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{\text{adp}}\uparrow$	$E_m^{\text{adp}}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{\text{adp}}\uparrow$	$E_m^{\text{adp}}\uparrow$	$MAE\downarrow$
✓	✓	0.922	0.918	0.973	0.019	0.876	0.885	0.935	0.043	0.871	0.873	0.897	0.058	0.900	0.901	0.921	0.036	0.908	0.892	0.957	0.024
✓		0.909	0.912	0.959	0.023	0.878	0.885	0.934	0.043	0.865	0.869	0.900	0.061	0.897	0.900	0.925	0.037	0.906	0.894	0.956	0.025
	✓	0.911	0.917	0.950	0.020	0.869	0.877	0.929	0.047	0.870	0.871	0.895	0.059	0.898	0.900	0.920	0.037	0.906	0.891	0.956	0.024

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TABLE IV

COMPARISON OF EFFICIENCY BETWEEN STANDARD CROSS-ATTENTION AND OUR PROPOSED CAFM

Method	FLOPs (G) ↓	Params (M) ↓	Latency (ms) ↓	Memory (MB) ↓
Standard Cross-Attention	0.469	0.197	0.25	64.31
CAFM (Ours)	0.044	0.020	0.36	31.05

TABLE V

ABLATION ANALYSIS ON THE WINDOW SIZE WITHIN THE CAFM MODULE. WE COMPARE OUR DEFAULT SETTING (4 × 4) AGAINST SMALLER (2 × 2) AND LARGER (6 × 6) WINDOW CONFIGURATIONS. THE BEST RESULTS ARE HIGHLIGHTED IN RED

Window Size	DES				SIP				LFSD				NJU2K				NLPR				STERE				DUTLF-Depth			
	$S_m\uparrow$	$F_\beta^{adp}\uparrow$	$E_m^{adp}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{adp}\uparrow$	$E_m^{adp}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{adp}\uparrow$	$E_m^{adp}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{adp}\uparrow$	$E_m^{adp}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{adp}\uparrow$	$E_m^{adp}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{adp}\uparrow$	$E_m^{adp}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{adp}\uparrow$	$E_m^{adp}\uparrow$	$MAE\downarrow$
2 × 2	0.890	0.885	0.938	0.028	0.865	0.868	0.927	0.049	0.868	0.876	0.896	0.058	0.889	0.894	0.923	0.040	0.900	0.880	0.955	0.026	0.880	0.874	0.920	0.044	0.900	0.908	0.947	0.038
4 × 4 (Ours)	0.922	0.918	0.973	0.019	0.876	0.885	0.935	0.043	0.871	0.873	0.897	0.058	0.900	0.901	0.921	0.036	0.908	0.892	0.957	0.024	0.884	0.873	0.922	0.042	0.911	0.917	0.950	0.034
6 × 6	0.894	0.883	0.945	0.027	0.861	0.858	0.923	0.051	0.860	0.863	0.882	0.065	0.887	0.889	0.914	0.041	0.900	0.875	0.954	0.027	0.877	0.864	0.913	0.045	0.892	0.891	0.938	0.042

re-weighted features. We define the per-pixel feature energy as the channel-averaged magnitude over the spatial dimensions (u, v): $E(x)(u, v) = \frac{1}{C}\sum_{c=1}^{C}|x_c(u, v)|$.

To measure the relative dominance of the depth branch, we compute the local depth energy ratio map:

$$r(u, v) = \frac{E(x^{dep})(u, v)}{E(x^{rgb})(u, v) + E(x^{dep})(u, v) + \epsilon}. \tag{23}$$

Here, ϵ (set to 10^{-8}) is a small constant added for numerical stability to avoid division by zero.

Let R be the global average of $r(u, v)$ over the spatial dimensions (u, v).

The key diagnostic metric is the shift in contribution, defined as $\Delta R = R_{post} - R_{pre}$, where R_{post} and R_{pre} are computed using the output $(\tilde{x})$ and input (x) features, respectively. A negative ΔR signifies that the CMAM has effectively reduced the relative importance of the depth branch, serving as a quantitative proxy for modality suppression.

Note that R and ΔR serve as probing tools to visualize modality re-calibration rather than new benchmark metrics.

d) Robustness to Depth Degradation: To simulate sensor failures, we introduce two depth degradation modes:

(i) `constant(0)`, representing missing depth with no structural information;

(ii) `shuffled pixels`, which preserves global intensity statistics but destroys spatial geometry, approximating severe misalignment.

As illustrated in Fig. 9 (left), we analyze ΔR under these conditions. We observe that ΔR is negative across all settings, suggesting that the CMAM naturally tends to modulate high-frequency noise often found in depth maps.

Crucially, compared to the baseline (`none`), the magnitude of this reduction is significantly amplified under the `constant(0)` and `shuffled pixels` scenarios. This indicates that the mechanism successfully identifies unreliable depth cues—whether due to signal loss or structural mismatch—and adaptively performs stronger suppression on the depth branch to prevent feature contamination.

The accompanying visualizations (Fig. 9, right) further validate this: despite depth quality issues, the learned attention map remains focused on the salient object (e.g., the "RS" logo), demonstrating a robust fusion strategy rather than a fixed weighting scheme.

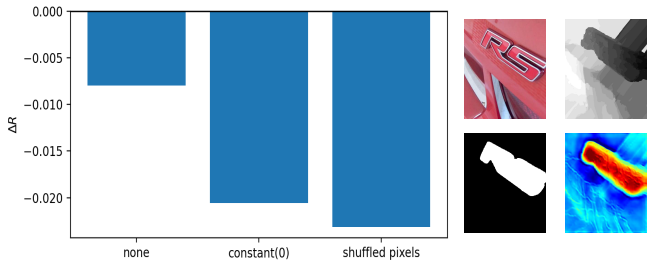


Fig. 9. Visualization of the dynamic modality re-calibration. The left bar chart displays the change in depth contribution ratio (ΔR) across three scenarios: normal input (`none`), zero-filled depth (`constant(0)`), and spatially shuffled depth (`shuffled pixels`). A larger negative value in ΔR indicates a stronger suppression of the depth branch. The right panel presents a representative qualitative sample (RGB, Depth, GT) alongside the feature activation map, highlighting the model's ability to maintain focus on salient regions despite potential depth imperfections.

3) ECM: To verify the effectiveness of the Edge Construction Module (ECM), we compared it with the Dual Edge Detection modules (DEDs) from DENet [9], a representative scribble-supervised model employing edge supervision. As shown in Table VI, although our ECM has a slightly larger parameter count due to its multi-scale design, it achieves significantly lower FLOPs (16.59G vs. 22.25G) and superior accuracy. This efficiency gain is attributed to ECM's ability to leverage low-resolution features from the Swin Transformer to construct high-quality edges, whereas DEDs typically requires high-resolution inputs. This demonstrates that our ECM is both effective and efficient in utilizing semantic information for edge construction.

As shown in Fig. 10, the edge feature maps extracted by our ECM are often clearer and cleaner compared to the DEDs, and can accurately distinguish between the foreground and background of the images. For example, in the first row detecting a doll, our module can directly identify the edges of the doll, while the DEDs also identifies the edges of the base. In the second row, we can identify the handlebars of the bicycle, whereas the DEDs fails to do so. This demonstrates the superiority of our module.

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TABLE VI

ABLATION STUDY RESULTS COMPARING ECM AND DEDS MODULES. ROWS REPRESENT MODULES, COLUMNS ARE GROUPED BY DATASET THEN METRIC. THE BEST RESULTS FOR EACH METRIC ON EACH DATASET ARE HIGHLIGHTED IN RED

Module	Overall		DES				SIP				LFSD				NJU2K				NLPR				STERE				DUTLF-Depth			
	FLOPS (G) ↓	Params (M) ↓	$S_m\uparrow$	$F_\beta^{adp}\uparrow$	$E_m^{adp}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{adp}\uparrow$	$E_m^{adp}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{adp}\uparrow$	$E_m^{adp}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{adp}\uparrow$	$E_m^{adp}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{adp}\uparrow$	$E_m^{adp}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{adp}\uparrow$	$E_m^{adp}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{adp}\uparrow$	$E_m^{adp}\uparrow$	$MAE\downarrow$
ECM	16.59	1.11	0.922	0.918	0.973	0.019	0.876	0.885	0.935	0.043	0.871	0.873	0.897	0.058	0.900	0.901	0.921	0.036	0.908	0.892	0.957	0.024	0.884	0.873	0.922	0.042	0.911	0.917	0.950	0.034
DEDs	22.25	0.22	0.919	0.910	0.973	0.020	0.870	0.877	0.930	0.047	0.865	0.868	0.900	0.061	0.893	0.893	0.916	0.039	0.904	0.882	0.955	0.025	0.876	0.863	0.915	0.046	0.905	0.905	0.946	0.037

TABLE VII

ABLATION STUDY QUANTIFYING THE EFFECTIVENESS OF THE NAMLAB [14] ALGORITHM. WE COMPARE OUR NAMLAB-BASED EDGE SUPERVISION AGAINST THE TRADITIONAL CANNY ALGORITHM AND TWO CNN-BASED EDGE DETECTORS (BDCN [71] AND HED [72]). NESSNET USING THE NAMLAB CONSISTENTLY ACHIEVES THE BEST PERFORMANCE ACROSS SEVEN BENCHMARK DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN RED

Method	DES				SIP				LFSD				NJU2K				NLPR				STERE				DUTLF-Depth			
	$S_m\uparrow$	$F_\beta^{adp}\uparrow$	$E_m^{adp}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{adp}\uparrow$	$E_m^{adp}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{adp}\uparrow$	$E_m^{adp}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{adp}\uparrow$	$E_m^{adp}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{adp}\uparrow$	$E_m^{adp}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{adp}\uparrow$	$E_m^{adp}\uparrow$	$MAE\downarrow$	$S_m\uparrow$	$F_\beta^{adp}\uparrow$	$E_m^{adp}\uparrow$	$MAE\downarrow$
Canny	0.907	0.902	0.954	0.023	0.874	0.876	0.932	0.044	0.863	0.863	0.892	0.061	0.894	0.894	0.915	0.039	0.905	0.885	0.957	0.024	0.880	0.867	0.921	0.044	0.906	0.903	0.945	0.037
BDCN	0.895	0.884	0.948	0.026	0.856	0.856	0.922	0.052	0.860	0.861	0.892	0.063	0.888	0.890	0.920	0.040	0.899	0.876	0.953	0.025	0.877	0.867	0.918	0.044	0.899	0.903	0.949	0.038
HED	0.905	0.895	0.966	0.024	0.866	0.870	0.934	0.047	0.861	0.860	0.905	0.063	0.890	0.891	0.922	0.039	0.900	0.877	0.954	0.026	0.877	0.867	0.919	0.044	0.899	0.900	0.944	0.038
NAMLab (Ours)	0.922	0.918	0.973	0.019	0.876	0.885	0.935	0.043	0.871	0.873	0.897	0.058	0.900	0.901	0.921	0.036	0.908	0.892	0.957	0.024	0.884	0.873	0.922	0.042	0.911	0.917	0.950	0.034

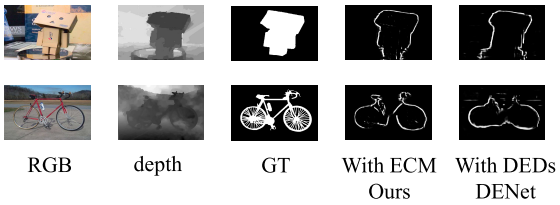


Fig. 10. The visual comparison results of the edge maps generated by the ECM module and the DEDs module.

4) NAMLab Edge Segmentation Algorithm: To rigorously quantify the novelty and effectiveness of our NAMLab-based edge supervision, we conducted a comprehensive ablation study comparing it against two categories of edge priors:

(i) traditional gradient-based methods, represented by the Canny algorithm (commonly used in weakly supervised domains [9]);

(ii) learning-based edge detectors, specifically the deep learning-based BDCN [71] and HED [72] networks.

We replaced the edge generation module in our pipeline with these alternatives while keeping other components unchanged.

As presented in Table VII, the NAMLab algorithm consistently outperforms both the Canny algorithm and the CNN-based counterparts (BDCN and HED) across almost all metrics on 7 datasets.

While CNN-based edge detectors provide high-quality boundaries, they often require heavy pre-training on external datasets and may not generalize as well to specific RGB-D salient object detection tasks without fine-tuning.

In contrast, our NAMLab approach demonstrates superior adaptability and effectiveness for foreground edge segmentation, providing more precise guidance for the model's edge stream without the need for additional heavy network branches.

E. Quality Analysis of Generated Pseudo-Labels

The efficacy of our second-stage refinement relies heavily on the quality of pseudo-labels generated by the first-stage model. Since the first stage is trained with sparse scribble supervision, a key concern is whether it can provide reliable dense supervision for the second stage, especially under noisy depth or complex scenarios.

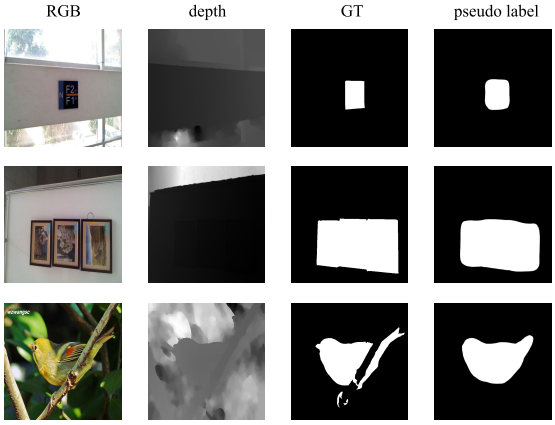


Fig. 11. Visualization of pseudo-label quality under noise.

Fig. 11 visualizes the pseudo-labels generated by the first-stage model.

1) Robustness to Interference: As shown in the first two rows of Fig. 11, the depth maps contain significant noise or lack valid geometric information.

However, thanks to the effective feature extraction and fusion (via the CMAM) in our network architecture, the first-stage model successfully suppresses these interferences and generates integral pseudo-labels that align well with the Ground Truth.

This confirms that our model does not blindly fuse modalities but selectively leverages valid information, providing a clean "Teacher" signal for the subsequent distillation stage.

2) Limitations: While the pseudo-labels accurately capture the main object structure (e.g., the bird body in the third row) amidst complex backgrounds, we observe that extremely thin structures (e.g., the bird's tail) are sometimes under-segmented.

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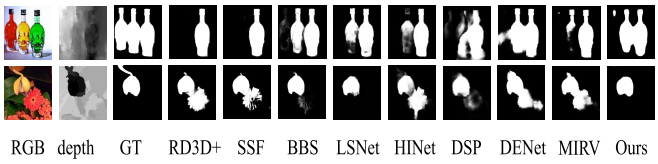


Fig. 12. Some failure cases of our NESS-Net.

This is a known limitation of scribble-supervision strategies, where boundary information is not explicitly provided during the initial training.

Nevertheless, these pseudo-labels provide a substantially denser and more informative supervision signal than the original sparse scribbles, facilitating the refinement in the second stage.

F. Failure Case

As shown in the first row of Fig. 12, our model has also failed to fully identify the leftmost bottle, recognizing only its cap. Furthermore, in the second row, our model has also failed to identify the top left stem of the yellow flower. This may be due to the poor quality of the depth images. As it can be seen from the corresponding depth images, the leftmost bottle is almost washed out by the white light and the flower stem is completely obscured by the shadow as well. In fact, it is quite common that salient objects cannot be distinguished due to poor quality of depth images. For example, in our provided visualization comparisons shown in Fig. 12, none of the remaining models can recognize the leftmost bottle, and similarly fail to identify the flower stem, sometimes even identifying noise entirely irrelevant to the salient object. However, overall, our model can always obtain the most stable results without introducing too much noise or bias in the predictions.

V. Conclusion

In this paper, we have proposed a novel scribble-supervised RGB-D SOD model called NESS-Net. To effectively guide the encoder in extracting and constructing the boundary features, we have presented a cascaded Edge Construction Module (ECM). To address the high complexity of the traditional cross-attention mechanisms, we have put forward a Cross-Attention Fusion Module (CAFM) based on the local window proposals and commonality suppression or enhancement. Based on the CAFM, we have also proposed a Cross-Modal multi-scale Attention-weighting Module (CMAM), which obtains quantified weights representing the importance of each modality through pyramid-like multi-scale feature extraction. Finally, the extensive experimental results on 9 benchmark RGB-D SOD datasets demonstrate that our methods and modules are effective in scribble supervision tasks as they outperforms all the current state-of-the-art scribble-supervised models and even achieves performance competitive to leading fully supervised models.

Our proposed NESS-Net for RGB-D SOD can be widely used in the fields of automatic driving, robot navigation, augmented reality, medical imaging analysis, and industrial quality assessment. Its principal innovation lies in achieving substantial reduction in annotation requirements while preserving competitive detection accuracy, rendering the NESS-Net particularly advantageous for real-world implementations characterized by data annotation challenges or demanding large-scale deployment scenarios.

References

[1] Y. Piao, W. Ji, J. Li, M. Zhang, and H. Lu, "Depth-induced multi-scale recurrent attention network for saliency detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 7254–7263.

[2] R. Ju, L. Ge, W. Geng, T. Ren, and G. Wu, "Depth saliency based on anisotropic center-surround difference," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2014, pp. 1115–1119.

[3] H. Peng, B. Li, W. Xiong, W. Hu, and R. Ji, "RGBD salient object detection: A benchmark and algorithms," in *Proc. Eur. Conf. Comput. Vis.*, 2014, pp. 92–109.

[4] L. Wang et al., "Learning to detect salient objects with image-level supervision," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 136–145.

[5] J. Li et al., "Joint semantic mining for weakly supervised RGB-D salient object detection," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 11945–11959.

[6] S. Gao et al., "Weakly supervised salient object detection algorithm based on bounding box annotation," *J. Comput. Appl.*, vol. 43, no. 6, pp. 1910–1918, 2023.

[7] J. Zhang, X. Yu, A. Li, P. Song, B. Liu, and Y. Dai, "Weakly-supervised salient object detection via scribble annotations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12543–12552.

[8] S. Yu, B. Zhang, J. Xiao, and E. G. Lim, "Structure-consistent weakly supervised salient object detection with local saliency coherence," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 4, pp. 3234–3242.

[9] Y. Xu, X. Yu, J. Zhang, L. Zhu, and D. Wang, "Weakly supervised RGB-D salient object detection with prediction consistency training and active scribble boosting," *IEEE Trans. Image Process.*, vol. 31, pp. 2148–2161, 2022.

[10] A. Li, Y. Mao, J. Zhang, and Y. Dai, "Mutual information regularization for weakly-supervised RGB-D salient object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 1, pp. 397–410, Jan. 2024.

[11] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[12] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. 3rd Intl. Conf. Learning Represent. (ICLR)*, 2015, pp. 1–14.

[13] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.

[14] Y. Zheng, B. Yang, and M. Sarem, "Hierarchical image segmentation based on nonsymmetry and anti-packing pattern representation model," *IEEE Trans. Image Process.*, vol. 30, pp. 2408–2421, 2021.

[15] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 9351. Cham, Switzerland: Springer, 2015, pp. 234–241.

[16] J. Fan, Q. Zhuo, and Y. Tang, "Attention-based region proposal network for few-shot object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 2340–2348.

[17] J. Li, Z. Wang, Z. Pan, Q. Liu, and D. Guo, "Looking at boundary: Siamese densely cooperative fusion for salient object detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 7, pp. 3580–3593, Jul. 2023.

[18] Y. Zhuge, Y. Zeng, and H. Lu, "Deep embedding features for salient object detection," in *Proc. AAAI Conf. Artif. Intell.*, Jul. 2019, vol. 33, no. 1, pp. 9340–9347.

[19] J.-J. Liu, Q. Hou, Z.-A. Liu, and M.-M. Cheng, "PoolNet+: Exploring the potential of pooling for salient object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 887–904, Jan. 2023.

[20] Y. Jiang, X. Li, K. Fu, and Q. Zhao, "Transformer-based light field salient object detection and its application to autofocus," *IEEE Trans. Image Process.*, vol. 33, pp. 6647–6659, 2024.

Authorized licensed use limited to: Jiangnan University. Downloaded on November 03,2026 at 01:26:48 UTC from IEEE Xplore. Restrictions apply.

[21] X. Zhao et al., "Motion aware memory network for fast video salient object detection," *IEEE Trans. Image Process.*, vol. 33, pp. 709–721, 2024.

[22] J. Zhang et al., "RGB-D saliency detection via cascaded mutual information minimization," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 4338–4347.

[23] Z. Chen, R. Cong, Q. Xu, and Q. Huang, "DPANet: Depth potentiality-aware gated attention network for RGB-D salient object detection," *IEEE Trans. Image Process.*, vol. 30, pp. 7012–7024, 2021.

[24] S. Gao, H. Xing, W. Zhang, Y. Wang, Q. Guo, and W. Zhang, "Weakly supervised video salient object detection via point supervision," in *Proc. 30th ACM Int. Conf. Multimedia*, Oct. 2022, pp. 3656–3665.

[25] N. Wang and X. Gong, "Adaptive fusion for RGB-D salient object detection," *IEEE Access*, vol. 7, pp. 55277–55284, 2019.

[26] D.-P. Fan, Z. Lin, Z. Zhang, M. Zhu, and M.-M. Cheng, "Rethinking RGB-D salient object detection: Models, data sets, and large-scale benchmarks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 5, pp. 2075–2089, May 2021.

[27] Z. Wei, Y. He, and Z. Cai, "Enhanced object tracking by self-supervised auxiliary depth estimation learning," 2024, *arXiv:2405.14195*.

[28] J. Luo, Y. Nie, W. Ren, X. Cao, and M.-H. Yang, "Correcting optical aberration via depth-aware point spread functions," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5541–5555, Aug. 2024.

[29] K.-J. Hsu, Y.-Y. Lin, and Y.-Y. Chuang, "Weakly supervised salient object detection by learning a classifier-driven map generator," *IEEE Trans. Image Process.*, vol. 28, no. 11, pp. 5435–5449, Nov. 2019.

[30] X. Zheng, X. Tan, J. Zhou, L. Ma, and R. W. H. Lau, "Weakly-supervised saliency detection via salient object subitizing," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 11, pp. 4370–4380, Nov. 2021.

[31] L. Chen and Q. Hou, "Uncertainty-aware multi-source fusion for weakly supervised saliency detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Jun. 2023, pp. 8878–8887.

[32] J. Zhang, T. Zhang, Y. Daf, M. Harandi, and R. Hartley, "Deep unsupervised saliency detection: A multiple noisy labeling perspective," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 9029–9038.

[33] X. Qian, C. Wang, W. Wang, X. Yao, and G. Cheng, "Complete and invariant instance classifier refinement for weakly supervised object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2023, Art. no. 5627713.

[34] W. Zhao, J. Zhang, L. Li, N. Barnes, N. Liu, and J. Han, "Weakly supervised video salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 16826–16835.

[35] Y. Wang et al., "WBNet: Weakly-supervised salient object detection via scribble and pseudo-background priors," *Pattern Recognit.*, vol. 154, Oct. 2024, Art. no. 110579.

[36] Y. Zhuge, H. Gu, L. Zhang, J. Qi, and H. Lu, "Learning motion and temporal cues for unsupervised video object segmentation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 5, pp. 9084–9097, May 2025.

[37] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, and A. Torralba, "Learning deep features for discriminative localization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2921–2929.

[38] J. Zhang, Y. Dai, T. Zhang, M. Harandi, N. Barnes, and R. Hartley, "Learning saliency from single noisy labelling: A robust model fitting perspective," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 8, pp. 2866–2873, Aug. 2021.

[39] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 3–19.

[40] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[41] J. Wei, S. Wang, and Q. Huang, "F3Net: Fusion, feedback and focus for salient object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2019, pp. 12321–12328.

[42] L. R. Dice, "Measures of the amount of ecologic association between species," *Ecology*, vol. 26, no. 3, pp. 297–302, Jul. 1945.

[43] F. Milletari, N. Navab, and S.-A. Ahmadi, "V-Net: Fully convolutional neural networks for volumetric medical image segmentation," in *Proc. 4th Int. Conf. 3D Vis. (3DV)*, Oct. 2016, pp. 565–571.

[44] C. H. Sudre, W. Li, T. Vercauteren, S. Ourselin, and M. J. Cardoso, "Generalised dice overlap as a deep learning loss function for highly unbalanced segmentations," in *Proc. Deep Learn. Med. Image Anal. Multimodal Learn. Clin. Decis. Support*, 2017, pp. 240–248.

[45] D.-P. Fan, M.-M. Cheng, Y. Liu, T. Li, and A. Borji, "Structure-measure: A new way to evaluate foreground maps," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 4558–4567.

[46] D.-P. Fan, C. Gong, Y. Cao, B. Ren, M.-M. Cheng, and A. Borji, "Enhanced-alignment measure for binary foreground map evaluation," in *Proc. Int. Joint Conf. Artif. Intell.*, Jul. 2018, pp. 698–704.

[47] C. J. van Rijsbergen, *Information Retrieval*, 2nd ed., London, U.K.: Butterworth, 1979, pp. 133–135.

[48] R. Achanta, S. Hemami, F. Estrada, and S. Susstrunk, "Frequency-tuned salient region detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 1597–1604.

[49] L. Li et al., "Robust perception and precise segmentation for scribble-supervised RGB-D saliency detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 1, pp. 479–496, Jan. 2024.

[50] H. Bi, R. Wu, Z. Liu, H. Zhu, C. Zhang, and T.-Z. Xiang, "Cross-modal hierarchical interaction network for RGB-D salient object detection," *Pattern Recognit.*, vol. 136, Apr. 2023, Art. no. 109194.

[51] Y. Cheng, H. Fu, X. Wei, J. Xiao, and X. Cao, "Depth enhanced saliency detection method," in *Proc. Int. Conf. Internet Multimedia Comput. Service*, Jul. 2014, pp. 23–27.

[52] G. Li and C. Zhu, "A three-pathway psychobiological framework of salient object detection using stereoscopic technology," in *Proc. IEEE Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2017, pp. 3008–3014.

[53] N. Liu, N. Zhang, L. Shao, and J. Han, "Learning selective mutual attention and contrast for RGB-D saliency detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 12, pp. 9026–9042, Dec. 2022.

[54] Y. Niu, Y. Geng, X. Li, and F. Liu, "Leveraging stereopsis for saliency analysis," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2012, pp. 454–461.

[55] P. Krähenbühl and V. Koltun, "Efficient inference in fully connected CRFs with Gaussian edge potentials," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2011, pp. 109–117.

[56] M. Zhang, W. Ren, Y. Piao, Z. Rong, and H. Lu, "Select, supplement and focus for RGB-D saliency detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 3469–3478.

[57] D.-P. Fan, Y. Zhai, A. Borji, J. Yang, and L. Shao, "BBS-Net: RGB-D salient object detection with a bifurcated backbone strategy network," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 275–292.

[58] J. Zhang et al., "Uncertainty inspired RGB-D saliency detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 9, pp. 5761–5779, Sep. 2022.

[59] N. Liu, N. Zhang, K. Wan, L. Shao, and J. Han, "Visual saliency transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 4702–4712.

[60] N. Liu, Z. Luo, N. Zhang, and J. Han, "VST+: Efficient and stronger visual saliency transformer," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 11, pp. 7300–7316, Nov. 2024.

[61] T. Hussain, A. Anwar, S. Anwar, L. Petersson, and S. Wook Baik, "Pyramidal attention for saliency detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 2877–2887.

[62] Q. Chen, Z. Liu, Y. Zhang, K. Fu, Q. Zhao, and H. Du, "RGB-D salient object detection via 3D convolutional neural networks," in *Proc. AAAI Conf. Artif. Intell.*, vol. 35, 2021, pp. 1063–1071.

[63] Q. Chen, Z. Zhang, Y. Lu, K. Fu, and Q. Zhao, "3-D convolutional neural networks for RGB-D salient object detection and beyond," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 3, pp. 4309–4323, Mar. 2024.

[64] W. Zhou, Y. Zhu, J. Lei, R. Yang, and L. Yu, "LSNet: Lightweight spatial boosting network for detecting salient objects in RGB-thermal images," *IEEE Trans. Image Process.*, vol. 32, pp. 1329–1340, 2023.

[65] H. Gao, Y. Su, F. Wang, and H. Li, "Heterogeneous fusion and integrity learning network for RGB-D salient object detection," *ACM Trans. Multimedia Comput. Commun. Appl.*, vol. 20, no. 7, pp. 1–24, Jul. 2024.

[66] G. Zong, X. Li, and Q. Xu, "Scenario potentiality-constrain network for RGB-D salient object detection," *Knowl.-Based Syst.*, vol. 310, Feb. 2025, Art. no. 112910.

[67] H. Gao, F. Wang, M. Wang, F. Sun, and H. Li, "Highly efficient RGB-D salient object detection with adaptive fusion and attention regulation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 4, pp. 3104–3118, Apr. 2025.

[68] J. Han, M. Wang, W. Wu, and J. Xu, "Perceptual localization and focus refinement network for RGB-D salient object detection," *Expert Syst. Appl.*, vol. 259, 2024, Art. no. 125278.

[69] Y. Su, H. Gao, M. Wang, and F. Wang, "Asymmetric cross-modality interaction network for RGB-D salient object detection," *Expert Syst. Appl.*, vol. 275, May 2025, Art. no. 127004.

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[70] G. Xing, M. Wang, F. Wang, F. Sun, and H. Li, "Lightweight edge-aware mamba-fusion network for weakly supervised salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5631813.
[71] J. He, S. Zhang, M. Yang, Y. Shan, and T. Huang, "BDCN: Bi-directional cascade network for perceptual edge detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 1, pp. 100–113, Jan. 2022.
[72] S. Xie and Z. Tu, "Holistically-nested edge detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1395–1403.



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